Multilingual Retrieval Fairness

Technical report

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- *Repository:** https://github.com/sujan1001/BDSRAE
- *Status:** Reproducible smoke pipeline and Phase-I design; final benchmark results are not claimed.

Abstract

This project evaluates whether a public-services retrieval assistant works equally well in English, Malayalam and Hindi. The central comparison is between lexical retrieval, multilingual dense encoders and hybrid search over a parallel document set. The study reports Recall@k, MRR, nDCG@10 and answer quality separately for each language, then measures the maximum-minus-minimum language gap. Mitigations include translation, hybrid ranking and an alignment-aware fine-tuning pilot. The system-design contribution is a reproducible data-to-evidence pipeline with a scale sweep, partition accounting, storage measurements and an optional Spark runner. The responsible-AI contribution is an explicit provenance register, language coverage audit, no-personal-data boundary, uncertainty reporting and a failure-analysis protocol.

1. Problem and research questions

Users often ask public-service questions in their strongest language. A system that performs well in English but retrieves incomplete or irrelevant evidence in Malayalam or Hindi can create unequal access even when the underlying service is nominally available. The project therefore treats multilingual retrieval as an empirical fairness problem rather than assuming that a multilingual model is equally reliable for every language.

The research questions are: (RQ1) what is the retrieval and answer-quality gap across the three languages for the same information need; (RQ2) which model and index configuration gives the best quality-cost trade-off; and (RQ3) which mitigation reduces the gap without unacceptable translation cost, latency, memory or index growth?

2. Data and governance

The repository's smoke test uses nine project-created synthetic documents and nine aligned queries covering senior pension, water-supply complaints and public-safety emergencies. Each record has a language tag, topic ID, answer key and gold document ID. This gives the pipeline a fully offline test fixture and ensures that no real personal data is committed. It is intentionally too small to support a substantive fairness claim.

The planned public-data audit records FLORES-200, MIRACL, Samanantar, AI4Bharat IndicCorp, IndicQA and Malayalam/Hindi Wikipedia in `data/source_registry.csv`. Exact release, license, attribution and coverage are checked before download. MIRACL is useful for languages and qrels it actually releases, but the Malayalam track must be supported by a separate documented source if MIRACL does not provide native Malayalam qrels. External raw data is kept out of Git unless redistribution rights are explicit.

3. Method and architecture

The pipeline has five stages. First, a source registry captures provenance, license status and language coverage. Second, Bronze JSONL is normalized into Silver records while retaining source IDs. Third, Gold query-document pairs are checked for leakage and aligned topic IDs. Fourth, BM25, dense and hybrid retrieval are run over language-specific candidates. Finally, the evaluator writes query-level metrics, language summaries, system metrics and a run manifest.

The smoke implementation uses a deterministic hashing encoder so that a clean checkout runs without network access or model downloads. Optional SentenceTransformers models can be enabled in the full environment;

requested and actual encoders are logged so a missing model cannot silently appear as a valid transformer result. Reciprocal-rank fusion combines BM25 and dense rankings. The translation pilot uses an English pivot query stored with each synthetic query. The alignment pilot adds an explicit topic feature and is labeled as a pilot rather than as full multilingual fine-tuning.

4. Evaluation protocol

For every query, the evaluator records the rank of the gold document, Recall@1/5/10, reciprocal rank, nDCG@10 and a smoke-test answer-quality indicator based on the top document's answer key. Results are grouped by language. For each metric, the fairness gap is the maximum language mean minus the minimum language mean, and a normal-approximation 95% confidence interval is included when there is more than one sample.

The systems evidence is generated by a scale sweep over 9, 90 and 900 rows in the smoke configuration. For each point, the runner records rows, logical partitions, serialized storage bytes, elapsed processing time and throughput. `scripts/run_spark_scale.py` is a separate optional runner that uses Spark when the full dependency is installed and otherwise writes an explicit not-run status. This separation keeps the smoke path reproducible while preserving a measurable distributed-processing path for the course requirement.

5. Mitigation and cost comparison

The baseline establishes the gap before mitigation. Hybrid retrieval is expected to improve robustness when lexical overlap is strong in one language and dense similarity is stronger in another. Translation can make a low-resource query more compatible with an English-heavy encoder, but it adds translation latency, possible meaning drift and a new failure surface. Fine-tuning should be attempted only after the benchmark and split are locked; the repository's alignment pilot exists to validate the evaluation path before GPU-intensive training.

Every mitigation is compared using the same language splits and query set. The result table should contain absolute metric values, gap reduction, elapsed time, index size, model size and translation tokens or calls. A mitigation is not considered successful if it improves the average while materially widening the worst-language gap or making the system too costly to reproduce.

6. Responsible-AI analysis

The system reports language-wise results rather than a single blended score. It preserves representative failures and distinguishes retrieval failure from answer-evaluator uncertainty. Human bilingual review is required for the final answer-quality sample because exact answer-key match can penalize valid paraphrases and multilingual automatic evaluators can be language-biased. The report must state which data track, model revision, split and uncertainty estimate supports every language score.

The project does not equate a small measured gap with universal fairness. The benchmark may omit dialects, code-switching, accessibility needs, or real public-service edge cases. The safe claim is limited to the evaluated data, task and model configuration. Any future public demo should display provenance and a limitation notice and should not accept real personal cases.

7. Reproducibility and deliverables

The repository provides `Dockerfile`, `docker-compose.yml`, pinned dependency ranges, a smoke configuration, pipeline scripts, a results notebook, report, model card, data card, source registry and a three-minute demo script. A clean checkout can run the smoke test offline with one command. The run manifest records the Python version, platform, Git commit, data validation, encoder fallback status and responsible-AI boundary.

The remaining Phase-II work is to verify the exact public dataset releases, build the aligned public-service document set, run the locked baseline and mitigation matrix, collect Spark evidence on a meaningful scale, conduct bilingual answer review, and replace this smoke report's status note with measured final results.

8. Acceptance criteria and handoff plan

The project is ready for a final benchmark run only when the data registry contains a checked license and coverage entry for every downloaded source, the parallel split has no topic leakage across train and test, and every query has a documented gold document or an explicitly recorded unanswerable label. The experiment configuration must be versioned before model comparison so that changes to language sampling, chunking, tokenizer, index parameters or qrels cannot be mistaken for a mitigation effect.

The minimum evidence package for each system is a query-level table, a per-language summary, a gap table, a scale table and a run manifest. The scale table must show at least one distributed-processing run with row count, partition count, elapsed time, throughput and serialized storage size. The responsible-AI package must include representative false positives and false negatives, the bilingual review protocol, the model/data card versions and a limitation statement. This makes quality, systems scale and governance auditable from the same commit.

The team handoff is intentionally split. Sujan owns the ingestion registry, normalization and Spark-scale evidence; Swasthik owns retrieval configurations, bilingual answer review and mitigation comparisons. Both members jointly lock the benchmark, review the fairness-gap interpretation, and approve the final README, notebook and three-minute demonstration. No result is reported as a final claim until both members can reproduce it from a clean checkout.

9. Metric definitions and interpretation

Recall@k is one when the gold document appears in the first k retrieved documents and zero otherwise. Reciprocal rank is the inverse rank of the first relevant document, or zero when no relevant document is retrieved. nDCG@10 uses the ranked relevance list at ten positions. The answer-quality smoke indicator is a deterministic exact-match check against the synthetic answer key; it is a pipeline check, not a substitute for human bilingual review. For each metric, the language gap is defined as the largest language mean minus the smallest language mean, while confidence intervals describe sampling uncertainty rather than proving fairness.

The preferred final comparison is therefore a compact quality-cost table: mean Recall@10, nDCG@10, answer quality, maximum-minus-minimum gap, query latency, index bytes, model bytes, translation calls and scale throughput. A mitigation should be described as beneficial only if its gap reduction is supported by the locked test set and its resource overhead is stated beside the gain. This interpretation prevents a single average score from hiding a large Malayalam or Hindi regression.

References

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